KSHITIJ ASHOK SAWARKAR

Forward Deployed Systems & Autonomous Robotics Engineer

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Portfolio | GitHub | LinkedIn | IEEE Xplore (Doc ID: 10627276)

Professional Summary

Forward Deployed Systems Engineer with an M.Tech in Control & Instrumentation from MNNIT Allahabad and a B.Tech in Electronics & Communication Engineering from RCOEM Nagpur. Combines low-level embedded hardware foundations with deterministic client-side runtimes, real-time computer vision, and industrial systems automation. Proven record in active manufacturing: architected production client-side schema engines executing under 50ms latency (npl-automation.pages.dev) that compressed 72-hour ERP reconciliation loops into sub-minute runs. Lead author of peer-reviewed edge-inference research in IEEE Xplore (Doc ID: 10627276). Experienced in authoring formal operational specifications (`AGENTS.md`), programming ROS2 nodes on physical mobile robots, and intercepting RF spectral signals via Software-Defined Radio.

Technical Skills

Forward Deployed & Runtimes: Cloudflare Pages, Client-Side Runtime Optimization, Deterministic Combinatorial Parsing, RESTful APIs, Git, Docker, Shell Scripting, Zero-Compute Architectures.
Autonomous Robotics & Embedded: ROS2 (composition nodes, lifecycle management), Embedded Linux, Microcontrollers (Raspberry Pi 5, Arduino Uno), Serial Protocols (CAN, UART, SPI, I2C), 2D LiDAR & Camera Synchronization, PWM Actuation.
Edge Perception & Signal Processing: Real-Time Vision (YOLOv5, OpenCV), Model Quantization (INT8), Software-Defined Radio (SDR), RF Signal Classification, Spectrogram Analysis (FFT, I/Q Streams), Scikit-Learn.
Cloud Infrastructure & Data: AWS (EC2, Lambda, S3, SageMaker, CloudWatch), Event-Driven Serverless ETL Pipelines, Vectorized Numerical Pipelines (NumPy, Pandas), Structured Schema Validation.

Professional Experience

Wakefit Innovations Ltd. *Bengaluru / Hosur, India*
Systems Automation Intern – PGET *Aug 2026 – Present*

- Architected and deployed an industrial client-side engine (npl-automation.pages.dev) on Cloudflare Pages, automating multi-tier Finished Goods (FG) code generation and New Product Launch (NPL) parameter trees across dynamic seater variants and colorways.
- Eliminated multi-day manual ERP spreadsheet reconciliation bottlenecks by implementing recursive client-side parsing algorithms, compressing end-to-end BOM derivation from 72+ operational hours into deterministic sub-minute runs with sub-50ms execution latency.
- Authored formal system constraints (`AGENTS.md`) and deterministic validation guardrails, preventing parameter collisions, duplicate SKU generation, and upstream ERP data ingestion errors across active manufacturing assembly lines.
- Engineered the pipeline to execute strictly client-side with 100% data isolation and zero server compute overhead, eliminating recurring cloud infrastructure maintenance costs while guaranteeing offline plant-floor readiness.
- Built and deployed operational micro-utilities including image-sequence-renamer.pages.dev to standardize SKU digital catalog pipelines for manufacturing teams.

AICTE Virtual Internship (AWS Academy) *Remote, India*
AWS Cloud Intern / AI Developer (Cert ID: 6d18deba5a7b05eac91dc74ba3592785) *May 2023 – Jul 2023*

- Architected the migration of local ML pipelines to AWS infrastructure, provisioning EC2 instances, security groups, and IAM policies for secure distributed processing.
- Engineered asynchronous, serverless ETL data pipelines using AWS Lambda triggered by Amazon S3 storage bucket events, automating ingestion, schema validation, and real-time preprocessing with zero idle compute costs.
- Trained and deployed hosted inference endpoints using AWS SageMaker (XGBoost, Linear Learner), tuning hyperparameter configurations for scalable real-time inference against production datasets.
- Configured CloudWatch metrics, automated threshold alerts, and autoscaling policies to maintain high endpoint availability while preventing resource waste.

UCN Cable Network Pvt. Ltd. *Nagpur, India*
Computer Networking Intern (Ref: UCN/SIP/2023) *Mar 2023 – Apr 2023*

- Configured enterprise routing hardware and managed Layer 2 and Layer 3 (L2/L3) network switches, establishing static and dynamic routing topologies across regional distribution points.
- Implemented IPv4 CIDR subnetting, VLAN segmentation, and DHCP/DNS server configurations to isolate operational traffic and optimize bandwidth allocation across mixed client networks.
- Executed physical Layer 1 and Layer 2 fault diagnosis using packet analyzers and cable-fault testers, eliminating signal attenuation, packet drops, and MTU mismatches across physical trunk lines.

RCOEM TBI Foundation Cell (Sharon Innovations) *Nagpur, India*
IoT Systems & Embedded Firmware Intern *Jan 2021 – Feb 2022*

- Developed embedded firmware and sensor interfacing routines across microcontrollers and IoT development boards for connected hardware products incubated under Sharon Innovations.
- Integrated serial protocols (UART, SPI, I2C) to bridge environmental transducers with edge communication modules, ensuring low-power telemetry transmission with zero packet loss.
- Built internal test automation scripts and Hardware-in-the-Loop (HIL) debugging routines to profile voltage stability, sensor drift, and signal noise across early prototype iterations prior to field trials.

Flagship Systems & Field Deployments

HiWonder MentorPi M1 Autonomous Platform | *ROS2 Humble, Raspberry Pi 5, OpenCV, Linux* *2026*

- Programmed and calibrated an Autonomous Mobile Robot (AMR) powered by a Raspberry Pi 5 single-board computer running ROS2 on embedded Linux (Hardware Telemetry).
- Configured modular publisher/subscriber nodes for real-time motor actuation, PWM driver loops, and telemetry streaming across a 4-wheel mecanum omnidirectional chassis.
- Interfaced onboard monocular camera feeds with OpenCV pipelines for visual marker tracking, target acquisition, and spatial orientation.

Real-Time Drone Detection & RF Signal Classification | *Python, SDR, RF Signal Processing, ML* *2025*

- Built an automated edge spectral sensing pipeline intercepting communication bursts between commercial UAV flight controllers and Ground Control Stations (GCS) across 2.4 GHz and 5.8 GHz ISM bands (Architecture Deck).
- Streamed raw I/Q signal data from SDR hardware, applying Fast Fourier Transforms (FFT) and spectrogram analysis to train lightweight neural classifiers distinguishing drone telemetry from ambient noise with sub-second latency.

Photovoltaic Sensorless Solar Tracker | *Embedded Control, Dynamic MPPT Tracking* *2026*

- Engineered a dual-axis tracking algorithm that eliminates external light-dependent resistors (LDRs) by repurposing the intrinsic photovoltaic I-V voltage curve of the solar panel as the diagnostic light sensor (Demonstration).

Gas Detection Safety System | *Arduino Uno, MQ-2 Sensor, Embedded C/C++* *2025*

- Designed an industrial gas-hazard detection circuit on Wokwi (Live Simulation), programming ADC threshold comparison and automatic alarm trip logic to trigger emergency alerts under volatile gas leak scenarios.

Publications & Research Pedigree

IEEE Xplore Conference Publication | *Lead Author* | Document ID: 10627276 *Nagpur, India*

- **"Automated Object Detection and Billing Architecture for Smart Retail Systems"**: Evaluated edge inference optimization, INT8 parameter quantization, and OpenCV processing pipelines deployed on resource-constrained ARM hardware under strict thermal and memory limits.

Education

Motilal Nehru National Institute of Technology (MNNIT) *Allahabad, India*
Master of Technology (M.Tech) in Control and Instrumentation *2025 – Present*

- Specialization in high-order dynamic systems, state-space modeling, multivariable feedback control, and real-time embedded signal processing.
- Serving as Teaching Assistant for undergraduate control systems lectures, tutorials, and laboratory sessions.
- Advanced IIT Certification: NPTEL "Introduction to Multi-Agent Systems".

Shri Ramdeobaba College of Engineering and Management (RCOEM) *Nagpur, India*
Bachelor of Technology (B.Tech) in Electronics and Communication Engineering *2020 – 2024*

- Core foundations in Microprocessors & Microcontrollers, Digital Signal Processing (DSP), RF & Microwave Systems, Embedded Hardware Design, and Communication Networks.

Verified Accreditations

Microchip Technology & AICTE NEAT: Embedded Developer Virtual Internship (Cert ID: 8e59ff18984afaa4dc3ad2d2e244dbd8)
AWS Academy & AICTE NEAT: AWS Cloud Virtual Internship (Cert ID: 6d18deba5a7b05eac91dc74ba3592785)